

Ex.no:6b	DATA VISUALIZATION USING POWER BI

AIM

To visualize and analyze the **California Housing Dataset** using Line Chart, Bar Chart, and Pie Chart with Power BI to understand housing value trends, ocean-proximity-wise comparisons, and housing distribution patterns.

ALGORITHM

Step 1: Import Pandas and load the California Housing Dataset into the working environment.

Step 2: Load the cleaned California Housing Dataset into Power BI Desktop.

Step 3: Group the housing data according to ocean_proximity and calculate the average median_house_value for each category.

Step 4: Plot the average house value using a **Line Chart** to observe variations across different ocean proximity categories.

Step 5: Group the dataset according to ocean_proximity and calculate the total number of housing records for each category.

Step 6: Plot the number of houses using a **Bar Chart** to compare housing records across different ocean proximity categories.

Step 7: Select the ocean_proximity categories and calculate their housing-record proportions.

Step 8: Plot the housing distribution using a **Pie Chart**.

Step 9: Add suitable titles, axis labels, legends, and display the visualizations.

CODE

```
# =====  
  
# EX NO: 6B  
  
# DATA VISUALIZATION USING POWER BI  
  
# CALIFORNIA HOUSING DATASET  
  
# =====
```

```
import pandas as pd

import matplotlib.pyplot as plt


# Load California Housing Dataset

df = pd.read_csv(

    "/content/housing prices.csv",

    low_memory=False

)


print("Dataset Loaded Successfully")


# -----

# 1. Line Chart – Average House Value by Ocean Proximity

# -----


house_value = (

    df.groupby("ocean_proximity")["median_house_value"]

    .mean()

)


plt.figure(figsize=(10, 5))


plt.plot(

    house_value.index,

    house_value.values,

    marker="o"

)
```

```
plt.title("Average House Value by Ocean Proximity")
```

```
plt.xlabel("Ocean Proximity")
```

```
plt.ylabel("Average Median House Value")
```

```
plt.xticks(rotation=45)
```

```
plt.grid(True)
```

```
plt.tight_layout()
```

```
plt.show()
```

```
# -----
```

```
# 2. Bar Chart – Number of Houses by Ocean Proximity
```

```
# -----
```

```
house_count = (
```

```
    df.groupby("ocean_proximity")
```

```
    .size()
```

```
    .sort_values(ascending=False)
```

```
)
```

```
plt.figure(figsize=(12, 6))
```

```
plt.bar(
```

```
        house_count.index,  
        house_count.values  
    )  
  
plt.title("Number of Houses by Ocean Proximity")  
plt.xlabel("Ocean Proximity")  
plt.ylabel("Total Number of Houses")  
  
plt.xticks(  
    rotation=45,  
    ha="right"  
)  
  
plt.tight_layout()  
  
plt.show()
```

```
# -----
```

```
# 3. Pie Chart – Housing Distribution
```

```
# -----
```

```
housing_distribution = (  
    df["ocean_proximity"]  
    .value_counts()  
)
```

```
plt.figure(figsize=(8, 8))

plt.pie(
    housing_distribution.values,
    labels=housing_distribution.index,
    autopct="%1.1f%%"
)

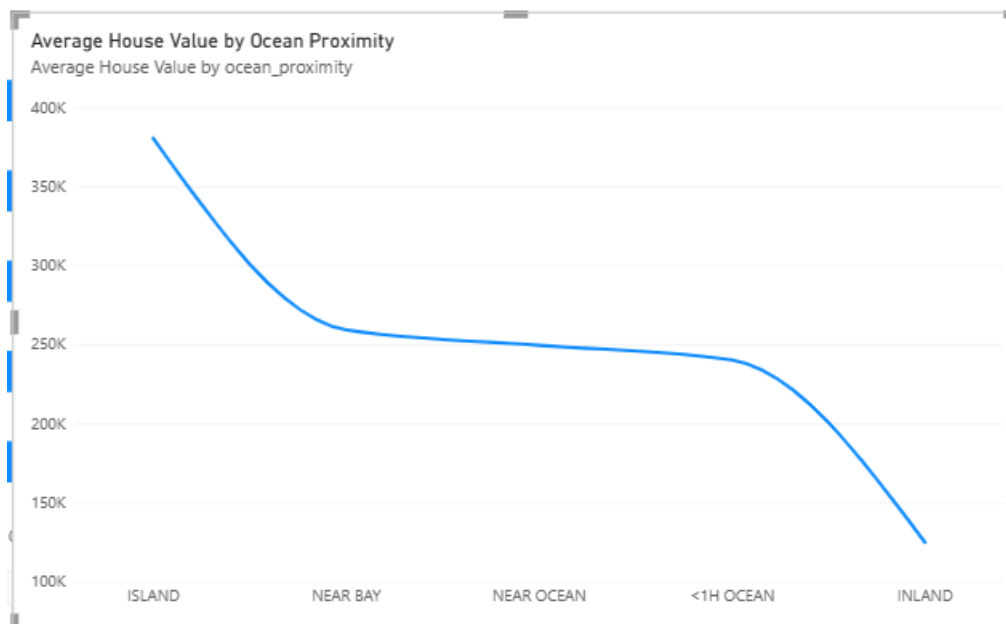
plt.title("Housing Distribution by Ocean Proximity")

plt.show()

print("\nEXPERIMENT 6B COMPLETED SUCCESSFULLY")
```

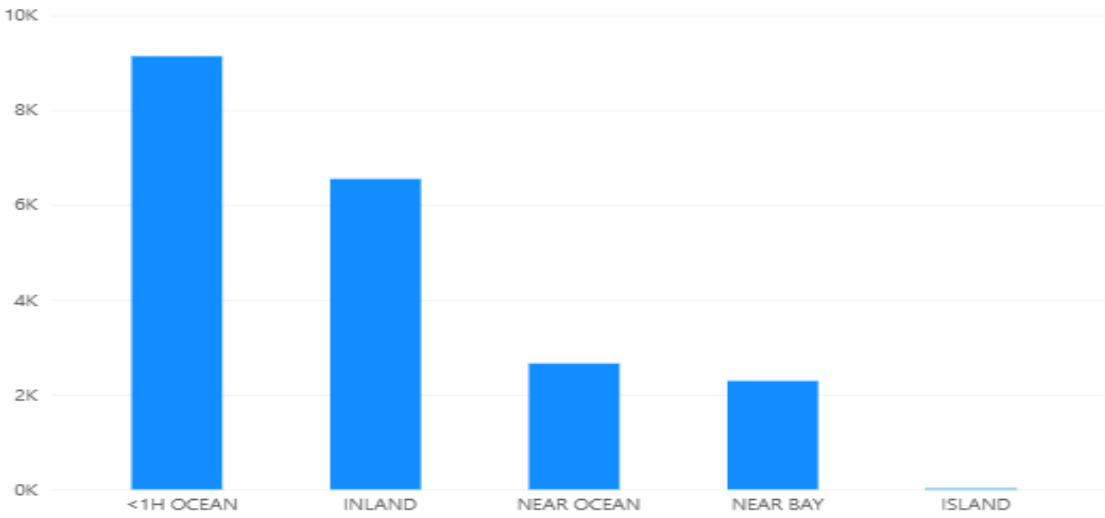
OUTPUT

LINE CHART



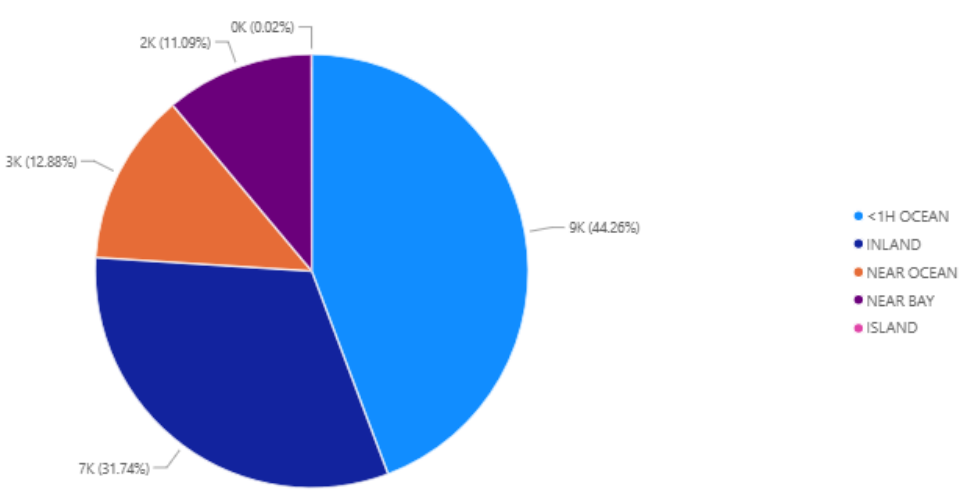
BAR CHART

Total Houses by Ocean Proximity
Count of ocean_proximity by ocean_proximity



PIE CHART

Total Houses1
by ocean_proximity



RESULT